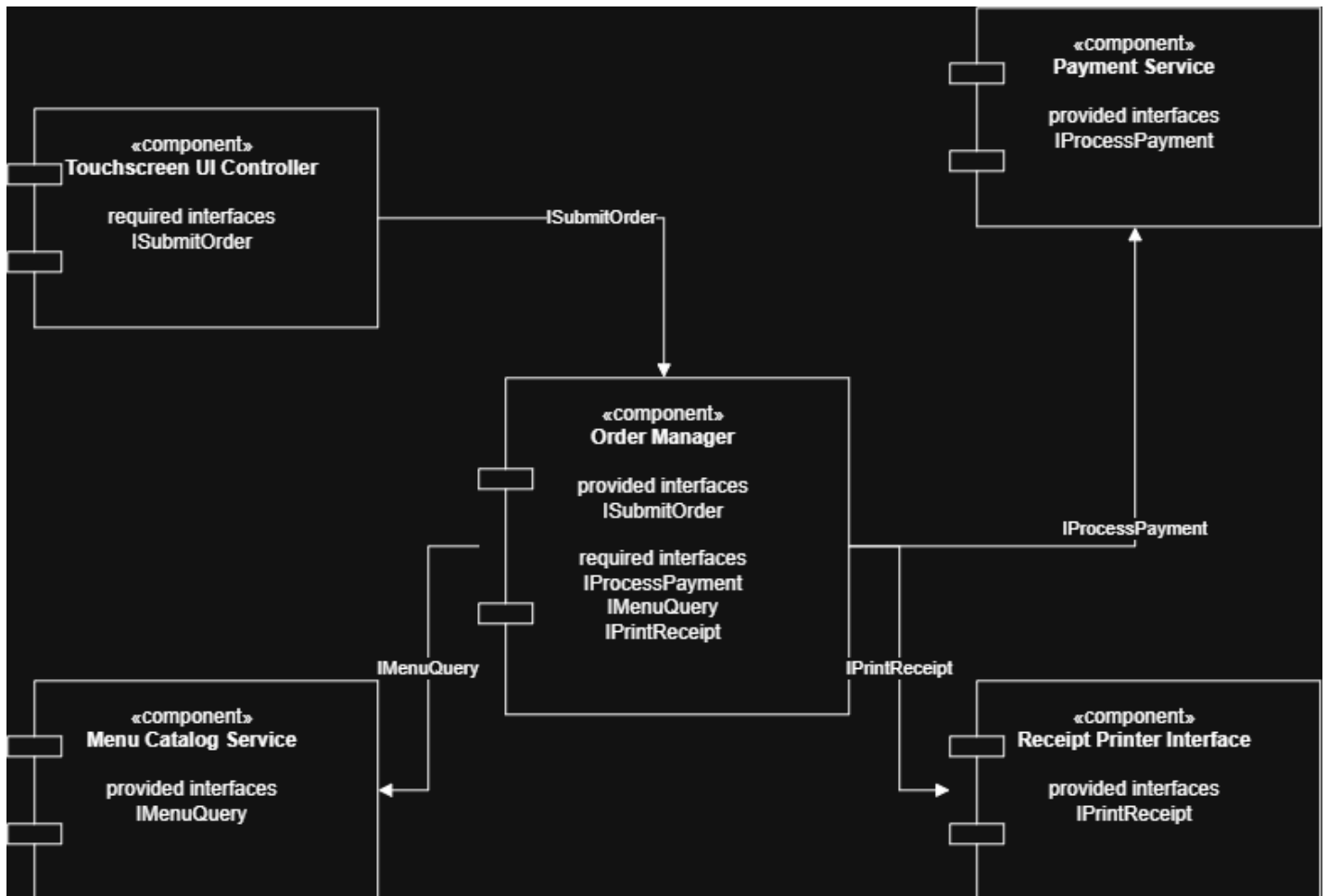


UML Component Diagram

Self-Service Coffee Kiosk System — Layered Architecture



Architectural Justification

Lab 3: Component Modelling & Architectural Pattern Selection | Self-Service Coffee Kiosk System

Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice:

Layered Architecture, organizing the system into a Presentation Layer (Touchscreen UI Controller), a Business Layer (Order Manager and Payment Service), and a Data & Hardware Layer (Menu Catalog Service and Receipt Printer Interface).

Reason 1 - Matches the kiosk's single-device, fixed-scope workflow:

The kiosk is one self-contained device with a small, fixed feature set: 3 coffee types, 2 sizes, and a single payment method (credit card). A Layered Architecture cleanly separates the touch-screen interaction logic (Presentation) from order and payment coordination (Business) and from menu/pricing storage and printer access (Data & Hardware), without the deployment and coordination overhead that Microservices would introduce for a system this small.

Reason 2 - Fits the sequential nature of the customer flow:

Every customer interaction follows the same sequence: browse the menu, select a coffee and size, pay, then print a receipt. This is a straightforward top-to-bottom flow, which a layered design handles naturally through the Order Manager coordinating calls down to Payment Service, Menu Catalog Service, and Receipt Printer Interface - far simpler to build and test than splitting this single flow across independently deployed services.

Security Advantage:

Card payment handling is isolated inside the Payment Service component in the Business Layer. The Touchscreen UI Controller in the Presentation Layer never talks to Payment Service directly - it only calls Order Manager through the ISubmitOrder interface, and Order Manager alone calls Payment Service through IProcessPayment. This keeps raw card-transaction handling out of the customer-facing layer, reducing the attack surface and narrowing the scope of components that must satisfy payment security requirements.

Performance Benefit:

Menu items and pricing are held in the Menu Catalog Service component in the Data & Hardware Layer and accessed by Order Manager through the IMenuQuery interface. Keeping this frequently-read data close to the business logic that needs it means browsing coffee types and sizes on the touch screen stays fast, independent of how long the payment step takes when Payment Service communicates with the external card network.